RESEARCH CONCEPT | NOT CLINICALLY VALIDATED

BreathSense X

Research-grade literature review and competition-prototype engineering report

Version 2.0 (revision of Revision 1.0, 17 October 2026) | Revision date: 18 October 2026
Indicative hardware envelope: INR 15,000

Proposed application: investigation of breast cancer-associated breath VOC patterns. Present deliverable: nonclinical sensing and reliability research platform.

How to read the evidence tags in this report

Tag	Meaning
[SUPPORTED IN V1]	Stated in the original Revision 1.0 report and its cited sources. Full-text re-verification is still pending (see Section 11).
[PROPOSED]	Design intent or method added in Version 2.0. Not yet built, measured or reviewed by a mentor.
[EXPERIMENTALLY UNVERIFIED]	A performance claim that can only be confirmed by the planned bench experiments.
[OPEN ITEM]	Information that could not be verified here. Left visibly open instead of being guessed (parts, prices, extra literature).
[FUTURE CLINICAL PHASE]	Belongs to a separate, ethics-approved clinical research stage. Not part of the V1 hardware.

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Executive summary

BreathSense X brings five researched concepts into one assessable platform: (1) ambient-reference measurements, (2) drift monitoring and recalibration flags, (3) sample-quality rejection, (4) multi-heater gas-sensing features, and (5) future clinical multimodal fusion. The novelty claim is a measurably improved low-cost integration, not the invention of electronic noses or diagnostic effectiveness. The strongest under-budget hardware arrangement uses one shared chamber, a switched inlet, a downstream pump, Bosch BME688 sensing, a Sensirion SGP40 auxiliary signal and environmental logging. **[SUPPORTED IN V1]**

> **Evidence boundary.** Breast-cancer breath literature reports encouraging classifications, but small cohorts, varying collection practices, controls, and weak external validation prevent direct transfer of those accuracy values to this design. Results must be measured, not assumed. [1-4]

What Version 2.0 changes compared with Revision 1.0

- The picture of a typical low-cost single-path e-nose is now included and labelled as a **simplified baseline**, not as a description of every commercial or research instrument (Section 3).
- A correct **single-chamber, two-source sequential** flow schematic, a valve truth table and a control state diagram replace the purely illustrative description (Section 4).
- Every purchased part now has a decision record with an explicit **OPEN ITEM** where the model, port behaviour or price is not yet verified. Pump, flow sensor and valve are marked **procurement NOT APPROVED** (Section 5).
- Clear statement that the BME688 heater is internal to the sensing element and that the chamber is not heated (Section 5.2).
- Testable specifications and traceability for all five innovations, a pre-planned experiment set E1 to E7, a claims ledger, and a go/no-go rule that falls back to a reliable sensor logger if the pneumatic parts cannot be verified (Sections 6 to 9).
- Final cost breakdown stays as the **very last section**, split into a Phase 1 logger and a Phase 2 pneumatic platform using the V1 planning allowances (Section 13).

Scope of this revision. Version 2.0 re-organises and tightens the V1 content and adds the engineering specification from the revision plan. It does **not** contain new experimental data, new price quotes, or a completed expanded literature search. Those items are listed as open in Section 12 so that nothing is invented.

1 | Problem description

1.1 Clinical motivation

Breast cancer screening and diagnosis use established clinical assessment, imaging and histopathology as indicated. A non-invasive breath test could someday become a complementary triage or research tool, but there is presently insufficient evidence for replacing those pathways. The challenge is not merely reading gases: it is differentiating a disease-associated signal from biological and environmental variability. **[SUPPORTED IN V1]**

Public-health context (incidence and burden figures) with a verified epidemiological reference: **[OPEN ITEM]** V1 did not cite one, so none is quoted here.

1.2 Breath biology

Exhaled air contains numerous volatile organic compounds (VOCs), influenced by oxidative stress, metabolism, medications, food, smoking, ambient exposures and other diseases. Compounds investigated in studies include aldehydes, ketones and hydrocarbons; there is no accepted single breast-cancer-specific breath biomarker. The 2022 systematic review identified 32 studies and reported substantial inconsistency between matrices and protocols. [1] **[SUPPORTED IN V1]**

1.3 Testable problem definition

Can an affordable electronic nose obtain reproducible gas-response signatures, identify compromised measurements, and reduce environmental and day-to-day drift on held-out nonclinical measurement sessions? Only after this instrument question is resolved should biomedical analytical and clinical partners assess whether breath profiles carry a robust breast-cancer-related signal. **[SUPPORTED IN V1]**

1.4 Feasibility objectives of Version 1 hardware

ID	Objective (nonclinical, measurable)	Linked innovation
O1	Estimate within-session, between-session and between-device (BME688 A vs B) variation of raw responses.	2, 4
O2	Test whether reference-aware features lower sensitivity to room-air and humidity changes on held-out days compared with sample-only features.	1
O3	Detect sensor drift across days using repeated reference cycles, with false-alarm and missed-drift rates.	2
O4	Evaluate a proposed quality gate: which faulty runs it rejects, at what coverage and residual error.	3
O5	Compare multiple supported heater profiles against a single profile for response quality, repeatability, power and time.	4

*All five objectives are **[PROPOSED]** and **[EXPERIMENTALLY UNVERIFIED]**. They do not involve patients and do not test any disease hypothesis.*

2 | Literature review and evidence synthesis

2.1 Search method and status

The V1 review is a curated, referenced feasibility review, not a registered systematic review and not a claim to exhaust all publications through October 2026. **[SUPPORTED IN V1]** Version 2.0 specifies the protocol below for the expanded review. The expanded search itself has **not** been executed in this document: **[OPEN ITEM]**

- **Review question:** which biological, sampling, sensing, correction and validation methods have credible evidence relevant to breast-cancer breath VOC research and robust e-nose measurement? **[PROPOSED]**
- **Databases:** PubMed, Google Scholar, IEEE Xplore, and Scopus / Web of Science where accessible. Record query strings, search date, years covered, inclusion and exclusion rules, and duplicates. **[PROPOSED]**
- **Target set:** about 15 to 25 relevant sources where available; state honestly if fewer are suitable. Do not call the review systematic without systematic-review methodology. **[PROPOSED]**
- **Source types kept separate:** peer-reviewed clinical evidence, preprints, conference abstracts, commercial marketing and manufacturer datasheets. Include contradictory and negative findings. **[PROPOSED]**
- **Topic buckets:** breast-cancer clinical studies; VOC biomarker reviews; GC-MS reference analytics; e-nose architecture; breath sampling; MOX humidity; sensor drift and domain shift; heater cycling; data leakage; clinical validation.

2.2 Core evidence carried over from V1

Evidence	What was studied	Finding / limitation
Leemans et al., 2022 [1]	Systematic review of 32 breast cancer VOC studies	VOC candidates are inconsistent across specimen types and measurement methods; supports standardized collection and careful validation.
Chemical Nose study, 2025 [2]	267 women (131 breast cancer, 136 healthy); Cyranose 320 and SVM	Reported 91% accuracy for its particular study design. Not a validated accuracy estimate for a BME688 system or general screening population.
JAMA Oncology review, 2018 [3]	63 publications across cancer breath research	Pooled sensitivity 79% and specificity 89%; methods, environment and patient physiology affect estimates.
Clinical Biochemistry meta-analysis, 2025 [4]	125 datasets, 8,768 cancer patients across cancer types	Pooled sensitivity 87%, specificity 81%; cross-cancer summary must not be treated as breast-specific device performance.

*All numbers above are quoted from V1 and are **[SUPPORTED IN V1]**. Exact author lists, titles, cohort sizes and metrics must be re-checked against the full text before release (Section 11).*

2.3 Evidence matrix (partially filled from V1)

The revision plan requires nine fields for every source. Only fields that V1 already states are filled; the rest are marked OPEN so nothing is guessed.

Reference	Design / participants	Instrument / handling	Reported result	Limits and relevance (what it does NOT prove)
[1] Leemans 2022	Systematic review, 32 studies. Controls: **[OPEN ITEM]**	Multiple platforms and matrices. Detail: **[OPEN ITEM]**	Inconsistent VOC candidates	Supports standardized collection. Does not identify a single reliable breast-cancer breath marker.
[2] Chemical Nose 2025	Case-control, 267 women (131 BC, 136 healthy). Benign controls: **[OPEN ITEM]**	Cyranose 320, 32 polymer sensors, collection bags, ambient-air purge, sample, recovery; SVM	91% accuracy (study-specific)	Shows an e-nose workflow. Not transferable to BME688; split method and CI: **[OPEN ITEM]**
[3] JAMA Oncol 2018	Review, 63 publications, many cancers	Mixed methods	Pooled sens 79%, spec 89%	Methods, environment and physiology influence estimates. Not breast-specific.
[4] Clin Biochem 2025	Meta-analysis, 125 datasets, 8,768 cancer patients	Mixed methods	Pooled sens 87%, spec 81%	Cross-cancer summary. Not breast-specific device performance.

2.4 What the best precedent teaches about workflow

The 2025 Cyranose study used a 32-polymer-sensor commercial e-nose, collection bags and a sequence of ambient-air purge, sample introduction and sensor recovery. It demonstrates an e-nose workflow, not a universally transferable medical classifier. [2] **[SUPPORTED IN V1]**

2.5 Synthesis questions for the expanded review

Question	Answer supported by V1 today	Status
Which VOC candidates recur and which fail to reproduce?	Candidates (aldehydes, ketones, hydrocarbons) vary across matrices and protocols; no accepted single marker [1].	**[SUPPORTED IN V1]** Detail: **[OPEN ITEM]**
What controls were used and how does that affect transferability?	Comparable benign breast-disease controls are lacking in many case-control datasets.	**[SUPPORTED IN V1]**
How many papers use patient-level or external-site separation?	External validation is limited overall. Fraction not quantified.	**[OPEN ITEM]**
Do studies measure chemistry or only pattern?	Sensor arrays recognise composite electrical patterns; GC-MS gives compound-resolved validation [1-3].	**[SUPPORTED IN V1]**
Water vapour, ageing and carryover?	Humidity effects, drift and carryover are listed as under-handled gaps.	**[SUPPORTED IN V1]** Quantification: **[OPEN ITEM]**
Which prior systems already have each of the five innovations?	Ambient-air purge and recovery exist in the Cyranose workflow [2]. Others not checked.	**[OPEN ITEM]**

2.6 Critical literature gaps

- Low reproducibility of named VOC biomarkers across analytical methods.
- Lack of sufficiently comparable benign breast-disease controls in many case-control datasets.
- Ambient-air and humidity effects inadequately handled or inconsistently reported.
- Sensor drift, between-device variation and potential carryover between measurement sessions.
- External validation, model calibration and realistic deployment-population testing are limited.

3 | Conventional breath e-nose and comparison baselines

3.1 How a conventional breath electronic nose works

Conventional sequence: patient/sample collection → transfer or inlet → pump/flow path → array of partially selective gas sensors → electrical readout → feature extraction → trained statistical classifier → experimental output. A separate ambient baseline or purge may be included, but architectures vary substantially. The sensor array recognizes composite electrical patterns rather than chemically identifying molecules. GC-MS and related analytical platforms provide compound-resolved research validation but are not interchangeable with inexpensive MOX sensors. [1-3] **[SUPPORTED IN V1]**

3.2 Simplified low-cost single-path baseline

Figure 1 shows the simplified baseline used in this report for comparison: one air path, constant flow, basic VOC sensors, basic temperature and humidity sensing and a basic controller. It is a teaching baseline, not a named product.

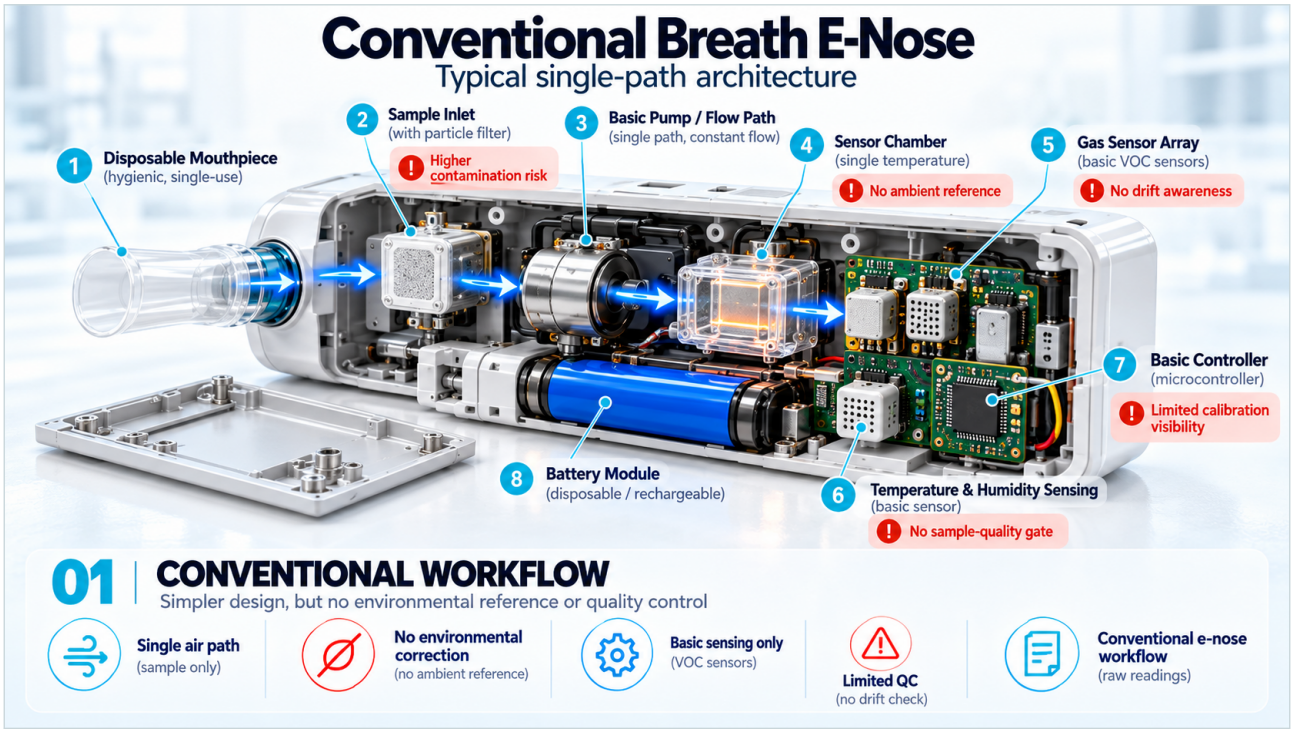


Figure 1. *Simplified conventional-style breath e-nose with a typical single-path architecture (illustrative concept visualization). The red annotations (no ambient reference, no drift awareness, no sample-quality gate, limited calibration visibility, higher contamination risk) describe* ***this simplified baseline only****. They are not claims about all commercial or research e-noses, several of which include reference air, purge and recovery cycles, compensation or machine learning (see Section 3.3). The image is not a specific instrument and not proof of any hardware performance.*

3.3 Comparison against real named workflows

Only features stated in the cited V1 sources are listed as present. Where a source does not confirm presence or absence, the cell is left OPEN instead of claiming absence.

Criterion	Simplified baseline (Fig. 1)	2025 Cyranose 320 study [2]	Analytical GC-MS workflow [1-3]	BreathSense X (proposed)
Air intake	One unreferenced intake with particle filter	Collection bags; ambient-air purge before sample introduction	**[OPEN ITEM]**	Switched ambient-reference and bench-sample sources with separate labels and logs **[PROPOSED]**
Detection	Basic VOC sensors	32 polymer sensors	Compound-resolved identification	2 x BME688 + 1 x SGP40 broad-response MOS; not compound identification
Control	Constant-flow pump, single temperature	Sequence of purge, sample introduction and sensor recovery	**[OPEN ITEM]**	Valve + downstream pump + logged purge/recovery **[EXPERIMENTALLY UNVERIFIED]**
Data	Raw readings	SVM classifier. Drift correction and QC reject option: **[OPEN ITEM]**	**[OPEN ITEM]**	Drift monitoring and quality gate with abstain/retest **[PROPOSED]**
Clinical	None	267 women (131 breast cancer, 136 healthy). Benign controls: **[OPEN ITEM]**	**[OPEN ITEM]**	None. Nonclinical air-condition experiments only **[FUTURE CLINICAL PHASE]** for patients
Product	Not applicable	Commercial instrument. Cost: **[OPEN ITEM]**	Laboratory instrument. Excluded from the INR 15,000 budget	Approx. INR 15,000 hardware envelope; open reproducibility **[PROPOSED]**

One additional well-described breath-research e-nose instrument is required by the plan: **[OPEN ITEM]**

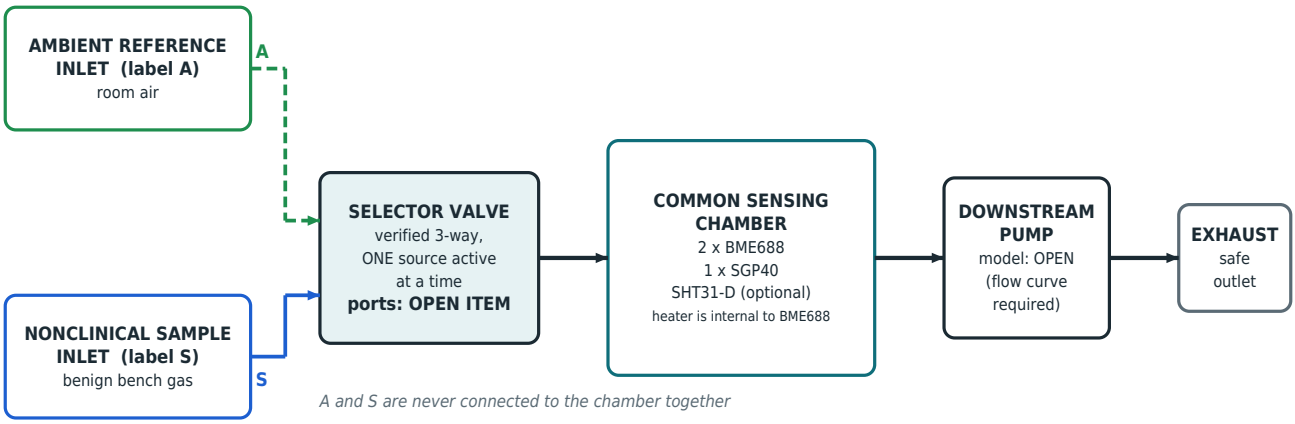
Prior-art overlap note. Ambient-air purge before sampling and sensor recovery already appear in the 2025 Cyranose workflow [2]. BreathSense X therefore does **not** claim to be the first to use a reference or purge. The claim is limited to a measurably improved, low-cost, open integration of reference, drift monitoring, quality gating and multi-heater features, to be shown by experiments. No claim is made that BreathSense X is more accurate than any existing device.

4 | Requirements and proposed BreathSense X architecture

4.1 Why one chamber instead of two parallel chambers?

A dedicated sample chamber and a separate reference chamber sound attractive, but their sensors will have manufacturing variability. At INR 15,000, a single switched chamber is more defensible for initial environmental-comparison experiments: ambient and sample streams visit the same sensors at different times. The trade-off is memory/carryover, switching delay and longer cycle time. Parallel chambers are a future option after matched-sensor characterization. **[SUPPORTED IN V1]** (design rationale, not measured)

4.2 Pneumatic schematic (sequential, single chamber, two sources)



Dashed line = ambient reference path (A). Solid line = sample path (S). Paths are identified by label, not by colour alone.

Flow direction left to right. Sources are selected sequentially, never mixed. Optional filters / flow meter: only after bench testing.

CONCEPTUAL P&ID - not a validated pneumatic or medical CAD drawing. Port layout, valve default state and pump placement need verification.

Figure 2. *Functional pneumatic schematic: isolated ambient inlet (A) and nonclinical sample inlet (S) → verified selector valve with one source active at a time → common sensing chamber → downstream pump → safe exhaust. Redrawn for Version 2.0 so that the two streams are never merged. Conceptual until the valve, pump and fittings are verified.*

4.3 Valve truth table and control logic

A product name alone does not establish that a valve has the needed port arrangement. The physical port layout and the normally selected (de-energised) state must be read from the datasheet and confirmed on the bench. **[OPEN ITEM]**

State	Selected inlet	Unselected inlet	Pump	Sensor read permission	Exit criterion / fault handling
BOOT	None (safe default; valve default per verified part: **[OPEN ITEM]**)	Not connected to chamber	OFF	Electrical check only	Sensors respond on I2C addresses, else FAULT
REFERENCE	Ambient (A)	Sample (S) must not reach chamber	ON	Yes, after settling	Stable baseline by pre-specified criterion (threshold: **[OPEN ITEM]**, set from pilot data)
PURGE / RECOVERY	Ambient (A)	Sample (S)	ON	Logged and flagged as transition; not used for features	Recovery criterion from pilot (**[OPEN ITEM]**); timeout leads to FAULT

State	Selected inlet	Unselected inlet	Pump	Sensor read permission	Exit criterion / fault handling
SAMPLE	Sample (S), benign nonclinical gas only	Ambient (A)	ON	Yes	Duration fixed from pilot (**[OPEN ITEM]**)
PURGE / RECOVERY (post)	Ambient (A)	Sample (S)	ON	Logged; used for carryover metric	Residual vs reference-only baseline within tolerance, else RETEST
QUALITY CHECK	Ambient (A)	Sample (S)	ON	Analysis only	Pass leads to SAVE; fail leads to RETEST or reject
FAULT	Verified safe default (**[OPEN ITEM]**)	-	OFF	Stopped, fault flag logged	Manual reset after inspection

4.4 Control state diagram

RETEST returns to REFERENCE. Green = reference-air state, blue = sample state (labels A / S in Figure 2).

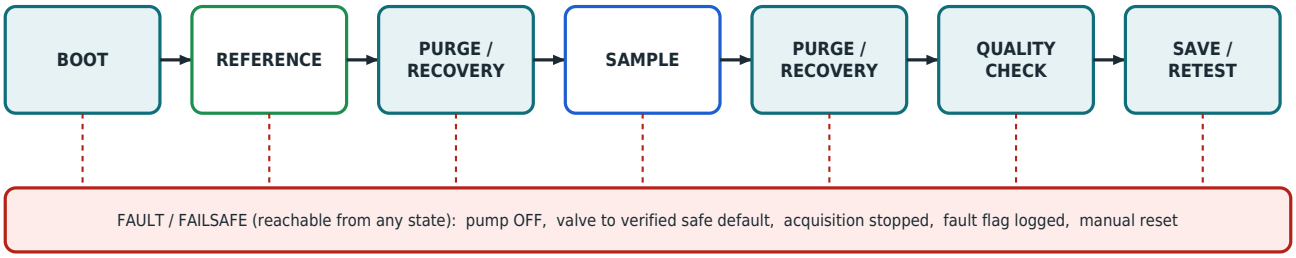


Figure 3. *Proposed control sequence BOOT → REFERENCE → PURGE/RECOVERY → SAMPLE → PURGE/RECOVERY → QUALITY CHECK → SAVE/RETEST with a failsafe state. Proposed logic, not yet implemented.*

4.5 Proposed measurement sequence

- **Power-on:** record device ID, firmware, heater profile and environmental conditions; verify stable electrical operation.
- **Ambient/reference phase:** select room-air source and record a reference response with timestamp.
- **Transition:** switch source; measure or verify purge and sensor recovery. Do not assume a fixed purge duration without experimental evidence.
- **Experimental sample phase:** feed only a safe, nonclinical gas or ambient-air comparison; collect raw response time series.
- **Post-measurement:** purge/recover while logging reference signals; discard runs with unacceptable carryover.
- **Data pipeline:** screen for unstable signals and instrument abnormalities; apply trained corrections; then compute research-only similarity/classification on approved engineering labels.

4.6 Cross-contamination / carryover strategy

Use separate labeled intake paths until the switching point; place pumping downstream of the sensing region when compatible with equipment; provide a controlled outlet/exhaust; minimize dead volume and use low-adsorption materials where practical; isolate reference and sample with a valve; log purge/recovery; perform blank-after-sample runs; inspect humidity condensation and tubing adsorption. A particle filter alone does not prevent VOC carryover, and filters may themselves adsorb target molecules. **[SUPPORTED IN V1]**

> **Safety boundary.** V1 is an ambient-air / benign environmental measurement demonstrator. It should not be used for patient sampling or clinical testing; any future human sampling requires ethics review, appropriate infection-control engineering, and supervised research protocols.

5 | Components: models, purpose, selection logic

5.1 Component table and verification status

Selection	Why it fits (V1)	Limit / evidence still needed	Procurement status
Bosch BME688 breakout x2 [5]	Gas resistance + temperature/humidity/pressure; configurable gas-heater operation; two boards enable between-device repeatability study.	Broad cross-sensitive response, not compound identification. Verify breakout actually carries BME688, board maker/revision, I2C addresses, raw resistance access, heater control interface, profile driver support, thermal placement.	Buy first, after board verification
Sensirion SGP40 breakout x1 [6]	Differently implemented VOC-sensitive raw signal; I2C; humidity-compensation support.	Indoor-air-quality design; VOC Index is not cancer evidence. Prefer raw SRAW and document compensation inputs, library and sample-rate limits. Independent value beyond BME688 to be shown.	Verify, then buy
ESP32-DevKitC / WROOM-32E compatible [7]	Low-cost controller with I2C, timing, serial/Wi-Fi logging and mature tools.	Does not replace PC machine learning. Select one exact retail board and revision; check 3.3 V logic, GPIO/I2C pinout, bus address, power draw and stable supply.	Buy first, after exact board choice
Sensirion SHT31-D breakout (optional)	Independent temperature/RH measurement near chamber.	Placement and independent calibration. Remove if no marginal benefit within budget.	Optional
FA0520E 12 V 3-way solenoid valve [8]	Affordable conceptual inlet-selection component for two air sources.	Stock uncertain (listing showed out-of-stock at V1 review). Must confirm true 3-port selector vs distributor behaviour, NC/NO state, leak rating, wetted materials, pressure/flow, 12 V current, gas compatibility.	NOT APPROVED: **[OPEN ITEM]**
12 V miniature diaphragm air pump, oil-free	Moves gas across the switched chamber; supports reference/recovery cycle.	Model TBD in V1. Need flow-pressure curve, wetted-material list, duty cycle, supply rating, noise, contamination profile. Prefer downstream placement if validated.	NOT APPROVED: **[OPEN ITEM]**
Inline flow sensor (upgrade)	Would verify actual flow rather than relying on pump speed.	Model TBD. Decide buy vs supervised external measurement. If omitted, V1 cannot verify exact gas flow and must say so.	NOT APPROVED: **[OPEN ITEM]**
PTFE tubing and compatible fittings	Relatively low-adsorption candidate fluid-path material; modular.	Check dimension/connector match, sealing, component contamination, leak tightness. Plan a test; do not assume no VOC loss.	Verify after valve/pump choice
USB power + protected driver electronics	Avoids battery integration during bench development.	Pump/valve need appropriately rated supply, MOSFET drivers, flyback suppression, fusing and supervised electrical review.	Verify with electrical review

5.2 Heater temperature versus chamber temperature

> **Important distinction.** The BME688 contains an internal micro-heater inside the sensing element. The heater temperature is a property of that element, not bulk heating of the sensing chamber, and the chamber is not intended to run at BME688 heater temperatures. The SGP40 is a separate sensor and must not be described as using the BME688 heater cycle. Heater profiles to be compared must be profiles that are documented and supported by the manufacturer; specific temperatures are **[OPEN ITEM]** until taken from the datasheet [5,6].

5.3 Component selection versus alternatives

BME688 vs MQ-series: prefer BME688 for controllable profiles, documented integration and digital data, not because it offers selective clinical VOC detection. **BME688 vs GC-MS:** a low-cost array can show pattern changes but cannot replace trace-compound analytical identification. **SGP40 vs another BME688:** SGP40 provides a different sensor signal, though both are broad-response MOS devices. **One versus two BME688:** two improve repeatability studies, not automatically independent gas features. [5,6] **[SUPPORTED IN V1]**

5.4 Mechanical and electrical integration considerations

Use separate stable power rails as required, appropriate switching drivers for pumps/solenoids, a shared signal ground where electrically suitable, short digital connections, a leak-checkable chamber, exhaust separation and physical distance between the BME688 heated element and temperature-sensitive electronics. Treat the conceptual exploded render as a visualization, not a build-certified mechanical specification. **[SUPPORTED IN V1]**

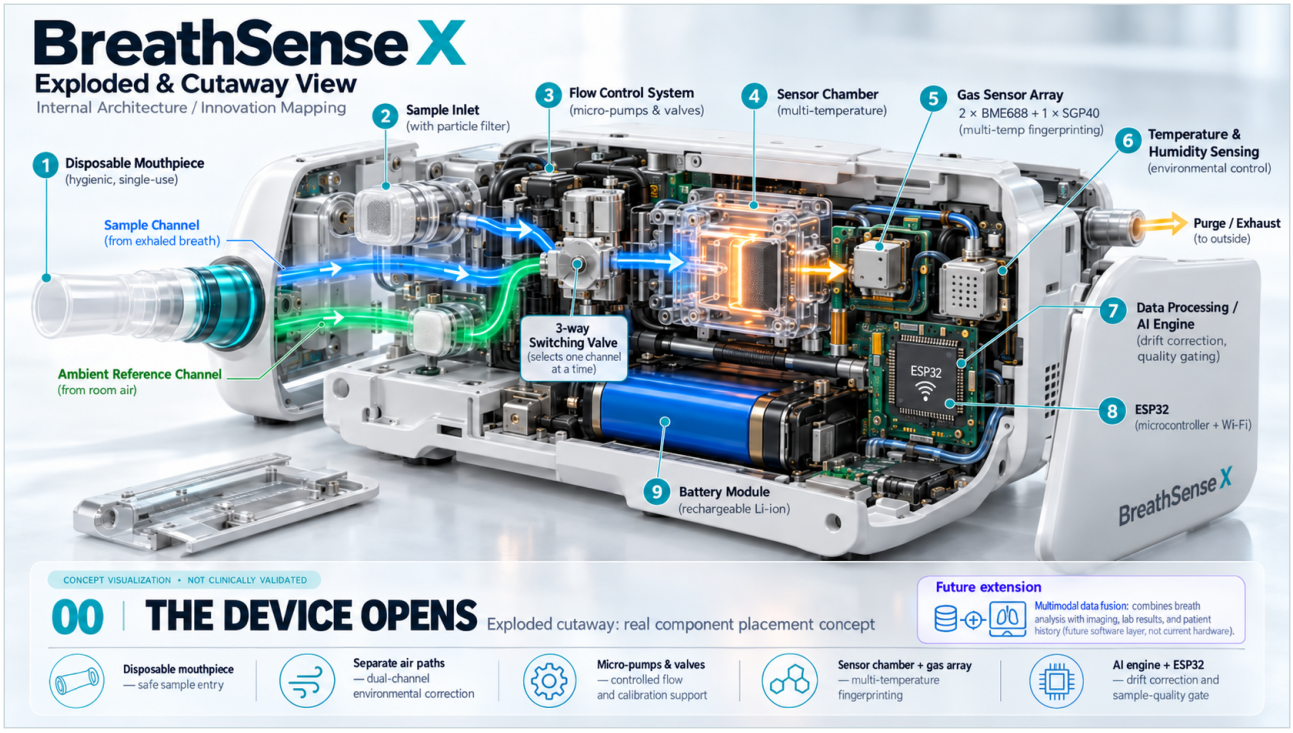


Figure 4. *BreathSense X exploded and cutaway view (concept visualization, not clinically validated).* ***Read with care:*** *the render is an illustrative concept. It is not proof of physical fit, engineering performance, safety, a validated bill of materials, or cancer detection, and it is not an assembled device. It shows a disposable mouthpiece, an exhaled-breath channel, a rechargeable Li-ion battery and an on-board "AI engine"; in V1 the build is USB-powered, nonclinical, and machine learning runs on a PC. The mouthpiece and battery are therefore future-concept elements, and the multimodal-fusion box is a future software layer, not V1 hardware.*

5.5 Design decision records and go / no-go rule

For each purchased part a decision record is required: required function → minimum specification → candidate model → manufacturer link → documented or measured suitability → competing model → rejection reason → supplier quotation date → expected delivery → verification task → remaining risk. Records for the valve, pump and flow sensor cannot be completed from the current sources: **[OPEN ITEM]**

> **Go / no-go.** If the exact valve and pump cannot be sourced and verified within budget, the full pneumatic platform is declared **Phase 2** and the deliverable becomes a reliable ESP32 + BME688 + SGP40 (+ SHT31-D) sensor logger with raw-data integrity (Phase 1). A passive 3D render must never be presented as a working fluidic subsystem.

6 | Mapping five innovations to hardware, evidence and tests

6.1 Hardware and evidence mapping (from V1)

Innovation	Physical / software enablers	What must be experimentally demonstrated
1. Environmental reference	Switched dual inlet + common chamber + humidity log	Lower held-out environmental sensitivity than sample-only baseline.
2. Drift-aware model	Repeated reference cycles + timestamp/device ID + correction estimator	Detect drift and improve stability on later unseen sessions without test-set tuning.
3. Quality gate	Baseline/noise/recovery checks + reject decision	Reject known poor-quality runs; report error and accepted fraction (coverage).
4. Multi-temperature fingerprint	BME688 firmware-supported heater-profile features	Independent improvement over simpler resistance features; monitor power/time.
5. Clinical fusion (future)	Data schema; external clinical imaging/history integration	Added predictive value beyond a clinical-only baseline in approved cohorts.

The fifth enhancement belongs to future clinical software development. It is not a physical module and must not consume the V1 hardware budget. ***[SUPPORTED IN V1]*** ***[FUTURE CLINICAL PHASE]***

6.2 Requirements traceability: gap to metric

Innovation	Literature gap	Components	Algorithm	Comparator	Metric and main limitation
1. Ambient reference	Ambient and humidity effects poorly handled [V1 gaps]	Selector valve, chamber, BME688, SHT31-D	Baseline-relative features	Raw sample response; humidity-only correction	Residual error and environmental sensitivity on held-out days, with CI. Reference may also harm: test it.
2. Drift awareness	Drift, between-device variation	BME688 A/B, SGP40, ESP32 clock and logs	Reference-cycle drift estimator and flag	No correction; flag only	False alarms, missed drift, post-correction stability on unseen days/devices. Must not hide sensor failure.
3. Quality gate	Carryover and unreliable runs	All sensors + valve/pump state logs	Transparent rules (missing data, range, unstable baseline, recovery, OOD)	Always accept	False reject rate, failure capture, coverage vs residual error. Thresholds from development data only.
4. Heater features	Single-resistance features may lose information	BME688 heater profiles	Profile summary features, simple models	Single supported profile	Held-out performance, repeatability, power, time. Heater stage is not chamber temperature.
5. Clinical fusion	No external validation; benign controls	None in V1 (software layer)	Clinical-only vs clinical-plus-breath	Clinical-only baseline	Future only; ethics-approved cohorts. No patient-facing output in V1.

Status of all five rows: ***[PROPOSED]*** ***[EXPERIMENTALLY UNVERIFIED]****. Prior-art overlap: ambient purge exists in [2]; overlap for drift, quality gating and heater cycling not yet checked:* ***[OPEN ITEM]***

7 | Data and machine-learning methodology

7.1 Logging schema and data dictionary

Record: timestamp, session ID, day ID, gas-source label, sensor and board ID, raw gas resistance, raw SGP40 signal, heater-profile phase where supported, temperature, RH, switch state, pump state, quality flags, firmware version and environmental notes. Raw files are immutable and carry a checksum or file hash. Never label engineered ambient-air conditions as cancer-positive or cancer-negative. **[SUPPORTED IN V1]** **[PROPOSED]**

7.2 Processing and model selection

- Establish baseline stability and sensor warm-up behaviour; retain raw values before transformations.
- Feature extraction: baseline-relative response, median/peak, slope, recovery, heater-phase summary, cross-sensor differences.
- Train correction using only training sessions; compare raw, humidity-corrected and reference-aware features.
- Benchmark ordinary linear models against Random Forest and SVM, not deep learning by default.
- Split by independent experimental session or chronological day; do not split temporally correlated readings randomly. Prefer held-out days and alternate-device evaluation for drift.
- Include humidity/temperature-only controls to diagnose environmental label leakage.
- Use a scikit-learn Pipeline (imputation where justified, scaling, feature extraction, environmental correction, classifier) fitted on training folds only. **[PROPOSED]**
- Predefine one primary metric per experiment and report paired differences with confidence intervals by session, not by correlated record.
- Report accuracy or ROC-AUC only where ground-truth class labels exist. For engineering tasks report variance, recovery, false reject rate and coverage.
- Keep a model card: intended use, forbidden medical use, data provenance, failure modes, evaluation limits and reproducibility steps. **[PROPOSED]**

7.3 Sample quality and drift rule design

V1 quality logic can use transparent engineering criteria: missing records, readings beyond supported range, unstable baseline, incomplete chamber recovery, instrument fault and out-of-distribution features. All threshold values must be established using development data and subsequently tested on held-out runs. Drift correction should never silently conceal sensor failure. Terms used in this report are “proposed quality gate” and “drift monitoring”; “self-calibrating” is avoided until reference standards and recalibration rules are validated. **[SUPPORTED IN V1]** **[EXPERIMENTALLY UNVERIFIED]**

8 | Validation plan and scientific acceptance criteria

8.1 Validation phases

Phase	Purpose	Content
A | Electrical and data integrity	Stable reads and traceable logs	Confirm stable sensor readings, power compatibility, switch-state logging, repeated acquisition; versioned hardware configuration and raw-data hash.
B | Instrument repeatability	Quantify variation	Nonclinical reference measurements over multiple independent days; within-session and between-session variation per sensor and feature; repeated readings inside one session are correlated.
C | Environmental correction ablation	Test the five innovations	Compare (A) raw single sensor, (B) fused sensors, (C) plus ambient reference, (D) plus drift correction, (E) plus quality rejection, on the same held-out sessions.
D | Carryover assessment	Measure memory effect	Compare a reference reading after a test condition with a reference-only baseline; report recovery time; never claim zero carryover without evidence.
E | Future clinical validation	Separate future stage	Ethics-approved recruitment, clinically verified diagnosis, benign breast-disease controls, independent-site testing, GC-MS where feasible, sample-size planning, sensitivity, specificity, calibration and net clinical benefit. **[FUTURE CLINICAL PHASE]**

8.2 Experimental protocol templates (benign, supervised, nonclinical)

ID	Design	Minimum documentation
E1 Warm-up and baseline	Repeated supervised benign-air sessions	Warm-up criterion, stabilisation plots, time of day and environment
E2 Valve isolation and leakage	Verify one selected branch only on an approved bench apparatus	Exact valve part, states, observed residual flow, uncertainty
E3 Purge / carryover	Reference → benign comparison → reference; repeated	Baseline shift vs no-comparison control; recovery criteria
E4 Humidity robustness	Naturally varying or safely supervised benign environments	Humidity/temperature confounding; no inferred chemistry
E5 Drift and sensor duplication	Later-day repeats of reference condition; BME688 A/B	Inter-device variation, drift errors, recalibration outcomes
E6 ML ablation	Matched dataset evaluated as simple vs enhanced models	Held-out evaluation, confidence intervals, coverage
E7 Reliability stress cases	Safe intentional metadata or signal-quality faults	QC sensitivity, false rejection, failure descriptions

*Acceptance thresholds must be justified by component specifications, pilot variability, the measurement purpose and mentor review. They are fixed **before** final testing and are never back-fitted after seeing final results. No threshold values are given here: **[OPEN ITEM]***

8.3 Engineering verification matrix

Requirement	Proposed bench evidence	Acceptance approach
Prevent mixing	Inspect valve state and flow paths under benign bench conditions	Quantify cross-port leak/isolation to an agreed, achievable threshold from part specifications
Limit carryover	Reference before and after separate benign conditions	Compare recovery residual with confidence interval; document baseline tolerance
Stable signals	Repeated reference sessions	Pre-specify allowable within-session variability using pilot measurements
Safe actuation	Verified power ratings and protective drivers	No overvoltage, uncontrolled heating or electrical faults
Data traceability	Session logs, metadata, firmware hashes	All key states logged; missing-data fraction documented

9 | Timeline, risks and claims ledger

9.1 Sixteen-week plan with go / no-go evidence

Period	Deliverable	Go / no-go evidence
Weeks 1-2	Literature matrix, finalized block design, datasheets	Requirements and safety review complete
Weeks 3-4	ESP32 + sensor acquisition demo	Time-stamped reliable raw logs
Weeks 5-7	Switched inlet and enclosed sensing assembly	Leak, switching and recovery behaviour documented
Weeks 8-10	Environmental and day-separated dataset	Repeated independent sessions with metadata
Weeks 11-13	ML ablations, quality gate, drift evaluation	Session-held-out comparisons and uncertainty
Weeks 14-16	Technical report, transparent demo and poster	Full BOM, reproducibility files, honest limitations

9.2 Risk register

Category	Risk	Mitigation
Technical	Sensor cross-sensitivity, humidity, inadequate limits of detection, condensation, contamination, valve leaks, heater effects, pump pulsation, dead volume, batch variation, electrical faults	Characterise on benign air first; leak and recovery tests; state limits openly; electrical review
Scientific	Temporal leakage, confounded classes, overfitting, no benign breast controls, poor probability calibration, selective reporting	Session-held-out splits, environment-only controls, pre-defined metrics, report all ablations
Operational	Valve stock, low-quality generic pumps, non-equivalent breakout boards, GST and shipping inflation, limited mentorship	Verify datasheets before ordering; reserve line in budget; fall back to Phase 1 logger
Ethical / claims	Mistaken medical claims	Analytical humility: limit claims to measured nonclinical performance; plan hospital and analytical collaborations separately

9.3 Claims ledger

Claim	Status
Breast-cancer breath studies report encouraging classification results [1-4]	**[SUPPORTED IN V1]** (full text to be re-checked)
The 2025 Cyranose study reports 91% accuracy in 267 women [2]	**[SUPPORTED IN V1]** (study-specific, not transferable)
A single switched chamber is more defensible than two chambers at this budget	**[PROPOSED]** design rationale; not measured
Ambient reference reduces environmental sensitivity	**[PROPOSED] [EXPERIMENTALLY UNVERIFIED]**
Drift monitoring flags drift and improves stability	**[PROPOSED] [EXPERIMENTALLY UNVERIFIED]**
Quality gate lowers residual error at a stated coverage	**[PROPOSED] [EXPERIMENTALLY UNVERIFIED]**
Multiple heater profiles improve over a single profile	**[PROPOSED] [EXPERIMENTALLY UNVERIFIED]**
BreathSense X detects or screens for breast cancer	**[UNSUPPORTED / PROHIBITED]**
BreathSense X is more accurate than existing e-noses	**[UNSUPPORTED / PROHIBITED]**
The exploded render shows a built, working or validated device	**[UNSUPPORTED / PROHIBITED]**

10 | Clinical translation plan (separate future stage)

Nothing in this section is part of the V1 hardware. **[FUTURE CLINICAL PHASE]**

- Ethics-approved recruitment with clinically verified diagnosis and benign breast-disease controls.
- Infection-control engineering for human sampling, and supervised research protocols.
- Analytical reference methods such as GC-MS where feasible, and independent-site testing.
- Biostatistical sample-size planning; evaluation of sensitivity, specificity, calibration and net clinical benefit.
- Multimodal fusion (imaging, laboratory results, patient history) evaluated against a clinical-only baseline, with datasets, permissions and endpoints specified in advance.
- The engineering phase does not justify patient-facing risk estimates.

11 | Core references and manufacturer documentation

[1] Leemans M et al. Volatile Organic Compounds Analysis as a Potential Novel Screening Tool for Breast Cancer: A Systematic Review (2022). https://pmc.ncbi.nlm.nih.gov/articles/PMC9134002/

[2] Chemical Nose-Based Non-Invasive Detection of Breast Cancer Using Exhaled Breath (2025). https://pmc.ncbi.nlm.nih.gov/articles/PMC11991366/

[3] Accuracy and Methodologic Challenges of VOC-Based Exhaled Breath Tests for Cancer Diagnosis. JAMA Oncology (2018). https://jamanetwork.com/journals/jamaoncology/fullarticle/2696723

[4] Diagnostic accuracy of breath VOC tests for cancer: systematic review and meta-analysis. Clinical Biochemistry (2025). https://doi.org/10.1016/j.clinbiochem.2025.110898

[5] Bosch Sensortec. BME688 product specifications and datasheet. https://www.bosch-sensortec.com/en/products/environmental-sensors/gas-sensors/bme688/

[6] Sensirion. SGP40 product and datasheet. https://sensirion.com/products/catalog/SGP40 and https://sensirion.com/resource/datasheet/sgp40

[7] Espressif ESP32 development boards. https://www.espressif.com/en/products/devkits

[8] Robu. FA0520E 12V miniature 3-way gas valve product listing. https://robu.in/product/3-way-small-mini-electric-solenoid-valve/

[9] Robu gas sensor catalogue (indicative local pricing). https://robu.in/product-category/gas-sensor/

[10] Robu ESP32 boards (indicative local pricing). https://robu.in/product-category/espressif-dev-boards-and-modules/

Source pricing represents observed listings and/or planning allowances, not guaranteed availability or an approved procurement quote. As of the V1 review, the cited FA0520E listing showed out-of-stock status. Version 2.0 did not re-fetch or re-verify any of these sources: **[OPEN ITEM]**

11.1 Source-to-claim table

Claim in this report	Source	Verification status
32 studies; inconsistent VOCs across matrices and protocols	[1]	From V1; full-text check pending
267 women (131 BC / 136 healthy), Cyranose 320, SVM, 91%, 32 polymer sensors, bags, purge-sample-recovery	[2]	From V1; full-text check pending
63 publications; pooled sensitivity 79%, specificity 89%	[3]	From V1; full-text check pending
125 datasets, 8,768 cancer patients; sensitivity 87%, specificity 81%	[4]	From V1; full-text check pending
BME688 capabilities; heater profiles and temperatures	[5]	Datasheet values: **[OPEN ITEM]**
SGP40 raw signal, humidity compensation, I2C	[6]	Datasheet values: **[OPEN ITEM]**
ESP32 board family	[7]	Exact board and revision: **[OPEN ITEM]**
Valve listing and stock status	[8]	Out of stock at V1 review; port layout: **[OPEN ITEM]**

12 | Revision log, glossary and unresolved verification tasks

12.1 Change log from Revision 1.0

V1 location	Change in Version 2.0
Cover, executive summary	Evidence tags added; scope of revision stated; unchanged scientific positioning.
pp. 2-3 Problem, literature	Feasibility objectives O1-O5, search protocol, partially filled evidence matrix, synthesis questions; unverified items marked OPEN.
p. 3 Conventional e-nose	New Figure 1; baseline relabelled as a simplified baseline; comparison against the Cyranose workflow and GC-MS; prior-art overlap note.
p. 4 Architecture	Schematic redrawn (sequential, no mixing); valve truth table and state diagram added.
p. 5 Components	Evidence-still-needed and procurement status added; pump, valve and flow sensor marked NOT APPROVED; heater clarification.
p. 6 Render and innovations	Figure 4 kept in the same place with a stronger illustrative-only caption; traceability table added.
p. 7 ML	Data dictionary, leakage rules, model card, metric rules added.
p. 8 Validation	Experiments E1-E7 and verification matrix added.
p. 8 Timeline, risks	Risk register and claims ledger added.
p. 9 Cost	Kept as the last section with Phase 1 / Phase 2 split and verification status.

12.2 Glossary

Term	Meaning
VOC	Volatile organic compound
MOX / MOS	Metal-oxide semiconductor gas sensor; broad, partially selective response
E-nose	Electronic nose: sensor array plus pattern recognition
GC-MS	Gas chromatography-mass spectrometry; compound-resolved analytical reference
BME688 / SGP40	Bosch gas/environment sensor with internal heater / Sensirion VOC sensor (raw signal SRAW)
RH	Relative humidity
I2C	Two-wire digital bus used by the sensors
P&ID;	Piping and instrumentation diagram
Carryover	Residual response from a previous gas that shifts a later baseline
Coverage	Fraction of runs accepted by the quality gate (not rejected or sent for retest)
SVM / AUC	Support vector machine / area under the ROC curve

12.3 Unresolved verification checklist

	Task
☐	Re-check every author name, cohort size, accuracy figure, date, title and DOI for [1]-[4] against the full text.
☐	Execute the expanded literature search (about 15-25 sources) and record queries, dates and inclusion logic.
☐	Add one more well-described breath e-nose instrument and verify which features it includes.
☐	Add a verified epidemiological reference for the clinical motivation.
☐	Confirm real board maker and revision of the BME688 and SGP40 breakouts, I2C addresses and library support.
☐	Select one exact ESP32 board and revision; check 3.3 V logic and power draw.
☐	Verify the valve port layout, NC/NO state, leak rating and wetted materials, or choose a validated alternative.
☐	Choose a pump with a published flow-pressure curve and decide the flow-sensor question.
☐	Take heater profiles and temperatures from the BME688 datasheet; do not invent values.
☐	Re-quote all prices with GST and shipping and note the quotation date and stock.
☐	Define acceptance thresholds from pilot data and mentor review before final testing.
☐	Complete the electrical review (drivers, flyback suppression, fusing, grounding) before powering pump and valve.

13 | Final cost breakdown (India; last section)

Engineering budget is calculated for a nonclinical, USB-powered switched single-chamber prototype. Costs are indicative INR planning amounts including approximate retailer/listing assumptions where available; verify the exact board model, taxes, shipping and stock before purchasing. **[SUPPORTED IN V1]** Prices were not re-verified for Version 2.0 (quotation date, GST and stock per item: **[OPEN ITEM]**).

Component / allowance	INR	Why included	Phase	Verification status
BME688 breakout x2	2,600	Primary gas resistance/profile + repeatability	1	Verify board; buy first
SGP40 breakout x1	1,100	Secondary raw VOC-sensitive signal	1	Verify raw SRAW access
ESP32 dev board x1	850	Data acquisition/controller	1	Choose exact board
SHT31-D environment breakout	500	Independent humidity/temperature	1	Optional
FA0520E or validated alternative 3-way valve	450	Select ambient vs sample path	2	NOT APPROVED; out of stock at V1 review
Mini diaphragm air pump	950	Move air through chamber (spec verify)	2	NOT APPROVED; model TBD
Pump/valve MOSFET drivers, suppression and power adapter	900	Controlled electrical actuation	2	Electrical review first
PTFE tube, fittings, connectors and exhaust tubing	1,200	Closed gas handling paths	2	Match to valve/pump ports
Chamber, seals, mounting and enclosure	1,600	Flow-path reproducibility / protected internals	2	Dimensioned design needed
PCB/protoboard, wires, headers, I2C mux if required	1,000	Electrical integration	1	Verify I2C address plan
Basic status screen/indicator	550	Demo feedback	1	Optional
Filters / material test blanks, lab consumables	800	Characterization / cleaning	2	Allowance
Reserve for shipping, spares, replacement	2,500	Uncertainty / contingencies	Reserve	Allowance
ESTIMATED TOTAL	**15,000**	**Target approximately INR 15,000**		

Phase split of the same INR 15,000 (arithmetic on V1 allowances)

Block	INR	Contents
Phase 1: sensor logger	6,600	BME688 x2, SGP40, ESP32, SHT31-D, protoboard/wiring, status screen. Reliable raw logger; always deliverable.
Phase 2: pneumatic platform	5,900	Valve, pump, drivers/power, PTFE and fittings, chamber/enclosure, consumables. Only after the valve and pump are verified.
Reserve	2,500	Shipping, spares, replacement, price and GST uncertainty.
Total	**15,000**	

Scope and exclusions. Total estimated: INR 15,000. This total is intentionally not a medical-device budget. GC-MS, calibration-gas services, clinical sampling, ethical approvals, institutional testing, and regulatory validation are excluded. Substitutions in pumps and valves may change costs and feasibility. A calibrated flow sensor may exceed the budget; if omitted, V1 cannot verify exact gas flow.

Purchasing priority. Buy the controller and BME688 modules first; verify robust digital readout before ordering gas-path hardware. Only source pumps, valves and tubes after documented compatibility and supervised electrical/mechanical review.